

SOLUTIONS - Exercise - Week 6: MATH1061/7861

(1) (4 marks)

(a) (2 marks) Use the Euclidean Algorithm to compute $\gcd(252, 198)$.

$$252 = 1 \times 198 + 54$$

$$198 = 3 \times 54 + 36$$

$$54 = 1 \times 36 + 18$$

$$36 = 2 \times 18 = 0$$

Therefore, we have $\gcd(252, 198) = 18$.

MARKING:

1 mark for correct algorithmic steps.

1 mark for identifying the GCD.

(b) (2 marks) Find the prime factorisation of 252 and 198.

$$252 = 2 \times 126 = 2 \times 2 \times 63 = 2 \times 2 \times 3 \times 21 = 2 \times 2 \times 3 \times 3 \times 7 = 2^2 \times 3^2 \times 7$$

$$198 = 2 \times 99 = 2 \times 3 \times 33 = 2 \times 3 \times 3 \times 11 = 2 \times 3^2 \times 11$$

Therefore, the prime factorisation of 252 is $2^2 \times 3^2 \times 7$ and the prime factorisation of 198 is $2 \times 3^2 \times 11$.

MARKING:

1 mark for correct prime factorisation of 252.

1 mark for correct prime factorisation of 198.

(2) **(6 marks)** The Fibonacci Sequence $F_1, F_2, F_3, \dots$ is defined by

$$F_1 = 1, \quad F_2 = 1, \quad F_{n+1} = F_n + F_{n-1} \quad \forall n \geq 2.$$

Use strong induction to prove that $F_n < 2^n$ for all positive integers n .

(1 mark) Base cases:

Notice that $F_1 = 1 < 2 = 2^1$ and $F_2 = 1 < 4 = 2^2$, so the statement is true for $n = 1$ and $n = 2$.

(2 marks) Inductive Assumption:

Suppose that $F_1 < 2^1, F_2 < 2^2, \dots, F_k < 2^k$ for some $k \geq 2$.

(1 mark) What we need to prove:

We want to show that $F_{k+1} < 2^{k+1}$.

(2 marks) Inductive Step:

Since $k \geq 2$, we have $k + 1 \geq 3$ and so

$$\begin{aligned} F_{k+1} &= F_k + F_{k-1} \\ &< 2^k + 2^{k-1} \quad (\text{As } F_k < 2^k \text{ and } F_{k-1} < 2^{k-1} \text{ by the inductive hypothesis}) \\ &< 2^k + 2 \cdot 2^{k-1} \\ &= 2^k + 2^k \\ &= 2 \cdot 2^k \\ &= 2^{k+1}. \end{aligned}$$

Therefore, by the principle of strong mathematical induction, $F_n < 2^n$ for all $n \geq 1$.

MARKING:

1 mark for correct base cases.

1 mark for the inductive statement.

1 mark for including the bound $k \geq 2$ in the inductive statement.

1 mark for stating what is needed to be proven in the inductive step.

2 marks for a valid proof of the inductive step.